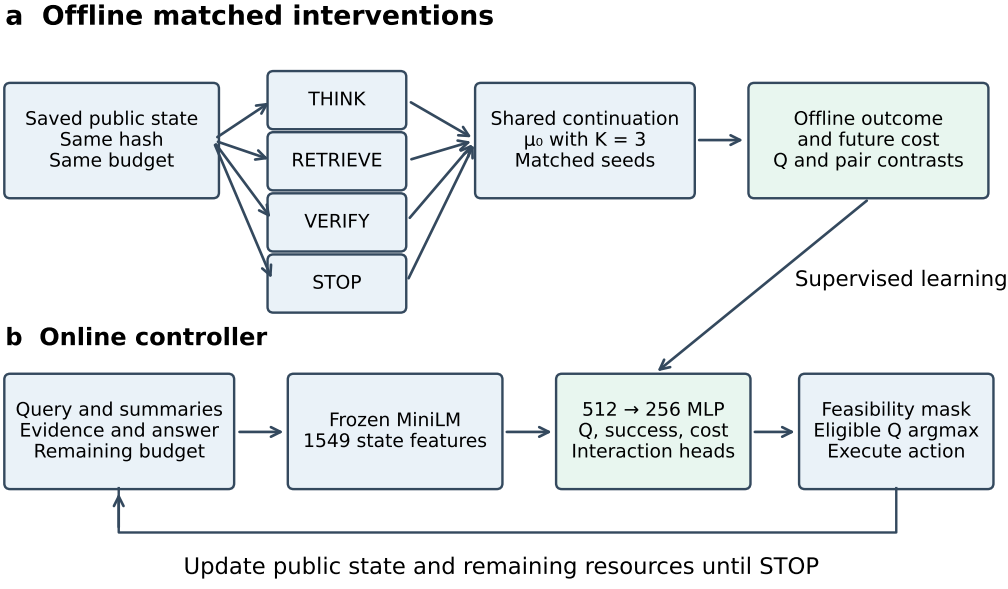
a Offline matched interventions
Saved public state
Same hash
Same budget
THINK
RETRIEVE
VERIFY
STOP
Shared continuation
μ0 with K = 3
Matched seeds
Offline outcome
and future cost
Q and pair contrasts
Supervised learning
b Online controller
Query and summaries
Evidence and answer
Remaining budget
Frozen MiniLM
1549 state features
512 → 256 MLP
Q, success, cost
Interaction heads
Feasibility mask
Eligible Q argmax
Execute action
Update public state and remaining resources until STOP